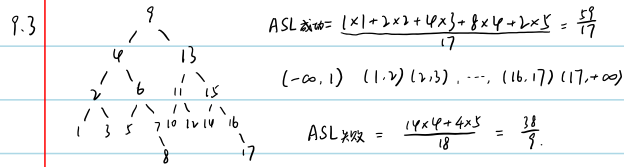
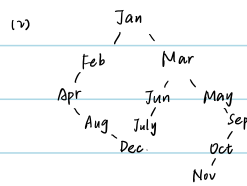


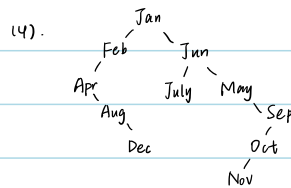
9.2 (1) 查 e: $e > d$ $e < f$ $e = e$ d.f.e
 (2) 查 f: $f > d$ $f = f$ d.f
 (3) 查 g: $g > d$ $g > f$ $g = g$ d.f.g



9.4 (1) ASL = $0.1 \times 1 + 0.25 \times 2 + \dots + 0.07 \times 12 = 5.43$



(3) ASL = $0.1 \times 1 + 0.25 \times 2 + 0.05 \times 3 + 0.01 \times 4 + 0.06 \times 5 + 0.11 \times 6 + 0.07 \times 7 + 0.02 \times 8 + 0.03 \times 9 + 0.1 \times 10 + 0.07 \times 11 = 3.2$



9.5

Key	10	25	33	19	06	49	37	76	60
H(Key)	10	3	0	8	6	5	4	10	5

(1)

i	10	3	0	8	6	5	4	1	7
次数	1	1	1	1	1	1	1	3	3
Key	10	25	33	19	06	49	37	76	60

即

i	0	1	2	3	4	5	6	7	8	9	10
Key	33	76	空	25	37	49	06	60	19	空	10

ASL_{成功} = $\frac{7 \times 1 + 2 \times 3}{9} = \frac{13}{9}$
 ASL_{失败} = $\frac{3 + 2 + 1 + 7 + 6 + \dots + 1 + 4}{11} = \frac{38}{11}$

(2)

i	10	3	0	8	6	5	4	2	9
次数	1	1	1	1	1	1	1	6	3
Key	10	25	33	19	06	49	37	76	60

即

i	0	1	2	3	4	5	6	7	8	9	10
Key	33	空	76	25	37	49	06	空	19	76	10

ASL_{成功} = $\frac{1 \times 7 + 6 \times 3}{9} = \frac{16}{9}$

(3)

i	0	1	2	3	4	5	6	7	8	9	10
Key	33	空	空	25	37	49	06	空	19	空	10

ASL_{成功} = $\frac{1 \times 7 + 2 \times 2}{9} = \frac{11}{9}$

9.7

```

bool Order (BiTree T, int *pre, bool *hasPre) {
    if (T == NULL) return true;
    if (!Order (T->lchild, pre, hasPre)) return false;
    if (*hasPre && T->key <= *pre) return false;
    *pre = T->key;
    *hasPre = true;
    return Order (T->rchild, pre, hasPre);
}
  
```

```

bool IsBST (BiTree T) {
    int pre = 0;
    bool hasPre = false;
    return Order (T, &pre, &hasPre);
}
  
```